

$$3d^2$$

July 22, 2026

$$1 \quad \left| 3d_{xz}^\alpha 3d_{xy}^\alpha 3d_{yz}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

Here are the CSFs:

$$|1, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

$$|2, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

$$|3, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle$$

$$|4, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|5, 1, 1\rangle = \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

$$|6, 1, 1\rangle = \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle$$

$$|7, 1, 1\rangle = \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|8, 1, 1\rangle = \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle$$

$$|9, 1, 1\rangle = \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|10, 1, 1\rangle = \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|11, 1, 0\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

$$|12, 1, 0\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

$$\begin{aligned}
|13, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|14, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|15, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|16, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|17, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|18, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|19, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|20, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|21, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|22, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|23, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|24, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|25, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|26, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|27, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|28, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|29, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|30, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|31, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|32, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|33, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|34, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|35, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|36, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|37, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|38, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|39, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|40, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|41, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|42, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|43, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|44, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|45, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 2 12 22

$$\begin{aligned}
\langle 2, 1, 1 | L_y | 1, 1, 1 \rangle &= I \\
\langle 2, 1, 1 | L_y | 8, 1, 1 \rangle &= -\sqrt{3}I \\
\langle 2, 1, 1 | L_y | 9, 1, 1 \rangle &= I
\end{aligned}$$

$$\Delta g_{yy}$$

$$\begin{aligned}
& -\zeta \Delta_{2-1}^{-1} \\
& -3\zeta \Delta_{2-8}^{-1} \\
& -\zeta \Delta_{2-9}^{-1}
\end{aligned}$$

$$\begin{aligned}
\langle 2, 1, 1 | L_z | 4, 1, 1 \rangle &= 2I \\
\langle 2, 1, 1 | L_z | 5, 1, 1 \rangle &= -I
\end{aligned}$$

$$\Delta g_{zz}$$

$$\begin{aligned}
& -4\zeta \Delta_{2-4}^{-1} \\
& -\zeta \Delta_{2-5}^{-1}
\end{aligned}$$

For a multiplicity of 3 we have the following:

$$D = \langle 1, 1 | 1, 1 \rangle - \langle 1, 0 | 1, 0 \rangle$$

$$E = \langle 1, -1 | 1, 1 \rangle$$

$$\begin{aligned}
\langle 1, 1 | \hat{H}_{eff} | 1, 1 \rangle &= -\zeta^2 (\\
& + \Delta_4^{-1} + \frac{1}{4} \Delta_5^{-1} + \frac{1}{8} \Delta_{11}^{-1} + \frac{3}{8} \Delta_{18}^{-1} + \frac{1}{8} \Delta_{19}^{-1} + \frac{1}{8} \Delta_{31}^{-1} \\
& + \frac{3}{8} \Delta_{38}^{-1} + \frac{1}{8} \Delta_{39}^{-1} + \frac{1}{4} \Delta_{41}^{-1} + \frac{1}{4} \Delta_{43}^{-1})
\end{aligned}$$

$$\begin{aligned}
\langle 1, 0 | \hat{H}_{eff} | 1, 0 \rangle &= -\zeta^2 (\\
& + \frac{1}{8} \Delta_1^{-1} + \frac{3}{8} \Delta_8^{-1} + \frac{1}{8} \Delta_9^{-1} + \frac{1}{8} \Delta_{21}^{-1} + \frac{3}{8} \Delta_{28}^{-1} + \frac{1}{8} \Delta_{29}^{-1} \\
& + \Delta_{34}^{-1} + \frac{1}{4} \Delta_{35}^{-1})
\end{aligned}$$

$$\begin{aligned}
\langle 1, -1 | \hat{H}_{eff} | 1, 1 \rangle &= -\zeta^2 (\\
& - \frac{1}{8} \Delta_{11}^{-1} - \frac{3}{8} \Delta_{18}^{-1} - \frac{1}{8} \Delta_{19}^{-1} + \frac{1}{8} \Delta_{31}^{-1} + \frac{3}{8} \Delta_{38}^{-1} \\
& + \frac{1}{8} \Delta_{39}^{-1} - \frac{1}{4} \Delta_{41}^{-1} - \frac{1}{4} \Delta_{43}^{-1})
\end{aligned}$$

$$\begin{aligned}
D = & \zeta^2 (\\
& + \frac{1}{8} \Delta_1^{-1} - \Delta_4^{-1} - \frac{1}{4} \Delta_5^{-1} + \frac{3}{8} \Delta_8^{-1} + \frac{1}{8} \Delta_9^{-1} - \frac{1}{8} \Delta_{11}^{-1} \\
& - \frac{3}{8} \Delta_{18}^{-1} - \frac{1}{8} \Delta_{19}^{-1} + \frac{1}{8} \Delta_{21}^{-1} + \frac{3}{8} \Delta_{28}^{-1} + \frac{1}{8} \Delta_{29}^{-1} - \frac{1}{8} \Delta_{31}^{-1} \\
& + \Delta_{34}^{-1} + \frac{1}{4} \Delta_{35}^{-1} - \frac{3}{8} \Delta_{38}^{-1} - \frac{1}{8} \Delta_{39}^{-1} - \frac{1}{4} \Delta_{41}^{-1} - \frac{1}{4} \Delta_{43}^{-1} \\
&)
\end{aligned}$$

$$\begin{aligned}
E = & \zeta^2 (\\
& + \frac{1}{8} \Delta_{11}^{-1} + \frac{3}{8} \Delta_{18}^{-1} + \frac{1}{8} \Delta_{19}^{-1} - \frac{1}{8} \Delta_{31}^{-1} - \frac{3}{8} \Delta_{38}^{-1} \\
& - \frac{1}{8} \Delta_{39}^{-1} + \frac{1}{4} \Delta_{41}^{-1} + \frac{1}{4} \Delta_{43}^{-1})
\end{aligned}$$